Freestream	
Alt (kft)	60
M	4
P (psf)	150
Pt (psf)	22738
T (R)	390
Tt (R)	1638
q (psf)	1677
Cdo	0.4
Ao (ft^2)	4.00
U (ft/s)	3872
md (sl/s)	3.47
An (ft^2)	0.4
Dn (lbf)	268
Dram (lbf)	13418

Constants	
gamma	1.40
Mwt_O2	32.00
Mwt_N2	28.16
Mwt_H2	2.00
Δhf	1.30E+09
a	8.80E-05
b	540

Chamber	
Tt1	540.0
Pt (psf)	200000
md ch	0.172
phi	2
beta	0.0
Cp_H2	87010
Cp_O2	5438
md H2	0.0343
md O2	0.1373
md N2	0.0000
m_brn H2	0.0172
Mbp	10.00
R (ft^2/s^2)	4972
Cp (ft^2/s^2)	17402
ht i (ft^2/s^2)	1.17E+07
η_b	59.6%
Ttb	5129

Throat	
A* (ft^2)	0.00633

Primary Exit	
Mx	2.35
Px	14794
Tx	2437.4
Ux	9679
Ae/A*	2.3
Ax (ft^2)	0.015
I (lbf)	1876
Htx (ftlb/s)	1.5E+07

Inlet	
Π_d	0.669
Pt (psf)	15212
Tt (R)	1638
md	3.47
phi	0.0
beta	3.762
md H2	0.0000
md O2	0.8038
md N2	2.6615
Mb	28.97
R (ft^2/s^2)	1716
Cp (ft^2/s^2)	6007

Station 15	
M	0.20
Pt (psf)	15212
P (psf)	14794
Tt (R)	1638
T (R)	1625
md (sl/s)	3.47
md H2	0.0000
md O2	0.8038
md N2	2.6615
Mbp	28.97
R (ft^2/s^2)	1716
Cp (ft^2/s^2)	6007
A (ft^2)	1.653
U (ft/s)	395
I (lbf)	25825
ht i (ft^2/s^2)	9.84E+06

Station 7	
md (sl/s)	3.64
phi	0.29
md H2	0.034
md O2	0.941
md N2	2.661
m_brn H2	0.034
Mb	27.45
R (ft^2/s^2)	1811
Cp (ft^2/s^2)	6339
ht i (ft^2/s^2)	9.93E+06
η_b	77.7%
Tt (R)	3071
Ht (ftlb/s)	7.1E+07
I (lbf)	27701
A (ft^2)	1.7E+00
rho (sl/ft^3)	2.72E-03
U (ft/s)	803
T (R)	3020
M	0.290
P (psf)	14860
Pt (psf)	15754

Nozzle	
Mb	27.45
R (ft^2/s^2)	1811
Cp (ft^2/s^2)	6339
Tt (R)	3071
T (R)	812
Pt (psf)	1.58E+04
P (psf)	150
M	3.73
A (ft^2)	6.67
U (ft/s)	5352
md (sl/s)	3.64
Cfg	0.96
F (lbf)	18687

Performance	
FN (lbf)	5000
SFCeng	3.98
FNWA	1443
ISP (s)	905